UMANG P. KAKLOTAR

COMPUTER VISION ENGINEER | AI & DEEP LEARNING DEVELOPER

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PROFESSIONAL SUMMARY

Computer Vision & Deep Learning Developer specializing in Automated Optical Inspection (AOI), real-time object detection, and industrial digital twin simulations. Hands-on experience developing and deploying custom Ultralytics YOLOv8 architectures across web image inputs, shop-floor mobile camera streams, and ROS 2 / Gazebo conveyor automation pipelines. Proficient in OpenCV, PyTorch, Python, FastAPI microservices, and multimodal AI platforms. Proven ability to architect end-to-end computer vision systems from dataset curation and model training to hardware-synchronized inference and operator dashboard analytics.

TECHNICAL SKILLS & CORE COMPETENCIES

Computer Vision & DL:	Ultralytics YOLOv8, YOLO, OpenCV, PyTorch, Torchvision, Scikit-Image, Object Detection, Bounding Box Annotation, Quality Classification, Defect Localization, Transfer Learning, Model Calibration.
Robotics & Simulation:	ROS 2 (Jazzy Jalisco), Gazebo Sim 8, Sensor Cameras, Industrial Conveyor Automation, CvBridge, Virtual Sensor Integration, Shop-floor Telemetry.
AI, Vector DB & LLMs:	Google Gemini Embedding API (gemini-embedding-2), FAISS Vector DB (IndexFlatIP), Groq LPU (Ultra-Low Latency Inference), LangChain, Multi-tenant RAG, Semantic Chunking.
Backend & Languages:	Python (FastAPI, NumPy, Pandas, Scikit-Learn, Uvicorn), JavaScript (ES6+), React 18, Vite, Node.js, HTML5, CSS3, REST APIs, Server-Sent Events (SSE).
Databases & Tools:	MongoDB, MySQL, Git, GitHub, Postman, Linux / Bash, VS Code, Clerk Auth.

FEATURED COMPUTER VISION & AI PROJECTS

PCB AI Defect Inspection & Smart Manufacturing System

FLAGSHIP INDUSTRIAL CV PROJECT

Tech: Ultralytics YOLOv8 (best.pt), OpenCV, ROS 2 Jazzy, Gazebo Sim 8, FastAPI, MongoDB, React | Live Demo: pcb-frontend.getvoroa.com

- **Trained Custom YOLOv8 Model:** Trained and evaluated custom YOLOv8 object detection model (best.pt) on PCB defect datasets to accurately identify 7 defect classes: Copper, Mousebite, Open, Pin-hole, Short, Spur, and Missing Hole with high confidence (>77% - 92%).
- **Multi-Surface Real-Time Pipeline:** Engineered dual-surface inspection pipeline processing high-res uploads and shop-floor mobile/desktop camera streams, generating real-time bounding boxes, defect count statistics, and automated Pass / Rework / Scrap decisions.
- **ROS 2 & Gazebo Digital Twin:** Built a virtual smart manufacturing inspection cell in Gazebo Sim 8 with ROS 2 conveyor automation; integrated virtual proximity sensors and virtual camera streams processed frame-by-frame via CvBridge into YOLOv8 for automated zero-contact inline defect detection.
- **Industrial Backend & Traceability:** Built high-performance asynchronous FastAPI REST endpoints with MongoDB for recording unique PCB IDs, defect coordinates, camera telemetry, and machine status for historical quality auditing.

Document AI — Enterprise Multi-Format RAG Platform

ENTERPRISE GEN-AI SYSTEM

Tech: Google Gemini Embedding-2, FAISS Vector Index, Groq LPU, LangChain, FastAPI, Clerk, React | Live Demo: rag-frontend-omega-three.vercel.app

- **Multi-Format Semantic Vector Ingestion:** Engineered document parsing pipeline for PDF, Word DOCX, CSV, TXT, and Markdown using overlap-aware recursive chunking (900 chars, 150 overlap); generated L2-normalized vectors via Gemini Embedding-2 for sub-millisecond similarity search using in-memory FAISS IndexFlatIP.
- **Ultra-Low Latency Inference & Citations:** Leveraged Groq LPU inference for real-time response token streaming via Server-Sent Events (SSE) with prompt guardrails to guarantee factual responses and verified citation bubbles showing source filenames and page numbers.
- **Multi-Tenant Architecture:** Implemented Clerk JWT authentication, maintaining per-user isolated vector spaces and encrypted MongoDB session chat history.

ADDITIONAL AI & MACHINE LEARNING PROJECTS

Real Estate Valuation & Home Price Prediction System

APPLIED MACHINE LEARNING

Tech: Python, Scikit-Learn, Pandas, NumPy, Matplotlib, Data Preprocessing

- **Data Cleansing & Feature Engineering:** Conducted Exploratory Data Analysis (EDA) on multi-variable housing datasets; applied automated outlier filtering, dimensional transformation, and categorical feature encoding.
- **Predictive Regression Modeling:** Trained linear and ensemble regression models; performed hyperparameter tuning and cross-validation to maximize accuracy and minimize Root Mean Squared Error (RMSE).

AI-Assisted Multi-Language Code Translation Engine

AI DEVELOPER TOOLING

Tech: React, Node.js, OpenAI API, AST Parsing concepts, Tailwind CSS

- **Syntax Mapping & Translation:** Constructed interactive web interface facilitating real-time translation between Python, JavaScript, and C++ with automated syntax validation and side-by-side code diff previews.

PROFESSIONAL EXPERIENCE

Frontend Developer Intern

6 MONTHS | 2025

Jemistry Info Solutions LLP — Surat, Gujarat, India

- **Modular Component Architecture:** Developed responsive, accessible web interfaces converting Figma designs into performant, clean frontend components using HTML5, CSS3, JavaScript, React, and Bootstrap.
- **REST API Integration & Performance:** Integrated RESTful APIs using Axios and Fetch API; implemented error boundaries, payload validation, and client-side caching to reduce redundant network round trips.
- **Version Control & Team Workflows:** Collaborated with agile engineering team using Git and GitHub for pull requests, code reviews, and resolving cross-browser rendering disparities.

EDUCATION

Bachelor of Computer Applications (BCA)

GRADUATING: 2025

Smt ZS Patel College of Management and Technology — Surat, Gujarat, India

- **Relevant Coursework:** Computer Vision Fundamentals, Machine Learning, Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Web Technologies, Operating Systems.

ENGINEERING METHODOLOGIES & CORE STRENGTHS

- **Automated Optical Inspection (AOI):** Inline camera synchronization, bounding box spatial IoU metrics, confidence thresholding, industrial quality classification (Pass / Rework / Scrap).
- **Embedded & Robotics Pipelines:** ROS 2 sensor_msgs / Image message handling, CvBridge zero-copy conversion, Gazebo SDF modeling, closed-loop conveyor station control.
- **Full-Stack AI Deployment:** FastAPI asynchronous microservices, Uvicorn worker scaling, WebSocket/SSE streaming, Docker-ready architectures, MongoDB schema design.